ADAPTIVE PREREGISTRATION FOR MODEL-BASED RESEARCH

A guide and template for preregistering ecological models

Table of contents

About this guide

This book/site provides a guide to implementing *Adaptive Preregistration for Model-Based Research*, with a focus on ecological modelling as it is used in both basic and applied contexts. This site covers the following topics:

1. Background on researcher degrees of freedom, the risk of questionable research practices (QRPs), and introduction to preregistration as a tool for mitigating QRPs and improving transparency.
2. Adaptive Preregistration
    1. *Overview*: What is adaptive preregistration, why do we need it for registering modelling analyses, and what are its defining features?
    2. *A practical guide*: How to implement adaptive preregistration using git and GitHub.
3. Preregistration template for model-based research in ecology and conservation.

This practical guide accompanies the paper:

> Gould, E., Jones, C. S., Yen, J. D. L., Fraser, H. S., Wootton, H. F., Good, M. K., Duncan, D. H., Hauser, C. E., Wintle, B. C., & Rumpff, L. (2026). 'But I can't preregister my research': Improving the reproducibility and transparency of ecology and conservation with adaptive preregistration for model-based research. *Methods in Ecology and Evolution*, *17*(6), 2041-210x.70311. https://doi.org/10.1111/2041-210x.70311

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Part I: Researcher Degrees of Freedom

From *Researcher Degrees of Freedom* to *Questionable Research Practices*

Researcher Degrees of Freedom

Throughout the course of model development and analysis researchers make numerous decisions. These alternative decisions are termed 'researcher degrees of freedom' (Simmons et al., 2011; Wicherts et al., 2016). These decisions are usually subjective and arbitrary yet methodologically and substantively acceptable (Wicherts et al., 2016), however may collectively significantly influence the final 'analysis strategy' (*sensu* Hoffmann et al., 2021) or 'specification' (*sensu* Simonsohn et al., 2020), and consequently the research findings and conclusions drawn from that analysis (Desbureaux, 2021).

Alternative decisions about model assumptions and implementation can lead researchers down very different analytical paths that result in 'parallel' or 'adjacent' worlds with different interpretations, and potentially conflicting conclusions originating from a single underlying dataset (Liu et al., 2020; Young & Stewart, 2021). The set of all (plausible or reasonable) possible decisions that a researcher could make over the entire analysis from beginning to end has been termed collectively as the 'garden of forking paths' (Gelman & Loken, 2013) or the 'multiverse', which consists of multiple universes of plausible analysis strategies (Steegen et al., 2016).

Researcher degrees of freedom are not inherently problematic, but rather are a product of the uncertainty that besets model development and analysis, whether due to incomplete theoretical understanding of underlying ecological processes, or because modelling is an act of simplification and abstraction; there is always some degree of choice about which ecological processes and phenomena should be included in the model, how they should be represented (Babel et al., 2019; Getz et al., 2017). Further uncertainty and multiplicity of analysis strategies stems from uncertainty in data pre-processing, model, parameter, and method uncertainty (Hoffmann et al., 2021).

Questionable Research Practices

When researchers are faced with ambiguous decisions where there is no single decision or strategy acceptable according to scientific standards and consensus, we are likely to select the decision that results in the most 'favourable' finding with "convincing self-justification" (Hoffmann et al., 2021; Simmons et al., 2011). Researchers routinely execute and explore alternative analysis pathways in service of identifying the most favourable (i.e. publishable) result, selectively reporting only one or several preferred results (Liu et al., 2020; Simmons et al., 2011).

Importantly, as Gelman and Loken note (Gelman & Loken, 2013) this behaviour is not a consciously nefarious 'fishing expedition', rather, because researchers are motivated actors in a 'publish or perish' culture who are susceptible to cognitive biases and are willing to change their modelling assumptions when the results conflict with desired outcomes – sincerely and self-convincingly believing that the more favourable model is superior (Young & Stewart, 2021).

Thus researcher degrees of freedom precipitate questionable research practices when they remain both undisclosed and exploited opportunistically (Bakker et al., 2018). Undisclosed and opportunistic use of researcher degrees of freedom artificially increases the probability of false positive results, inflates effect size estimates and ultimately reduces the replicability of published findings while encouraging overconfidence in the precision of results and exacerbating the risk of accepting and propagating false facts (Hoffmann et al., 2021; Nissen et al., 2016; Roettger, 2019; Wicherts et al., 2016). Emerging evidence suggests that rates of QRPs in ecology and evolutionary biology are similar to rates observed in other disciplines beset by reproducibility crises (Fraser et al., 2018).

Preregistration mitigates against researcher degrees of freedom

Preregistration is an open-science practice adopted in scientific disciplines that have begun to confront the 'reproducibility crisis', and aims to improve research transparency and mitigate questionable research practices within a study by distinguishing between what is a genuine a priori planned analysis, and what is not (Parker et al., 2019). Preregistration requires the methods and analysis plan of a future study are registered in a secure and publicly accessible platform prior to any data collection, and after submission it can no longer be altered.

Despite enthusiastic uptake of preregistration in some scientific disciplines, there has been little uptake of pre-registration in ecology and conservation, and many modellers and researchers engaged in non-hypothesis testing research both within and outside of ecology have eschewed preregistration on the grounds that preregistration and existing templates are irrelevant because they are focused on null-hypothesis significance testing (e.g. McDermott et al., 2021; Prosperi et al., 2019).

Instead of eschewing preregistration completely, this guide is premised on the idea that preregistration can and should be used in model-based research in ecology and conservation, but its particular form should reflect the norms and practice of the research context in which it is applied, in order to adequately restrict researcher degrees of freedom that may uniquely emerge within a domain-specific methodology. The work presented in this guide aims to translate preregistration into model-based research within ecology and conservation by proposing an expanded view of preregistration we term *Adaptive Preregistration*. We also present a template for preregistering model-based research in ecology and conservation.

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Part II: Adaptive Preregistration

A primer on *Adaptive Preregistration*

Background & Motivation

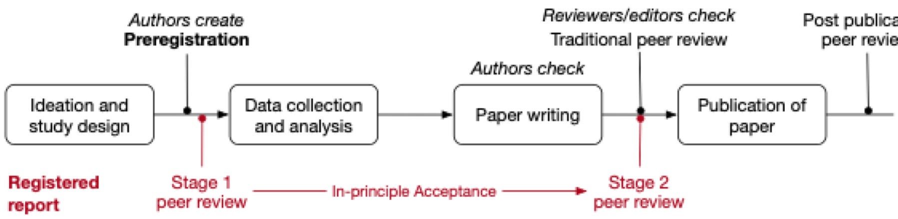

Figure 1: Preregistration as a component of the registered report publication pathway, as illustrated by (Pu et al., 2019).

The iterative nature of model development is a significant barrier to adopting preregistration for model-based research. Translating the preregistration process to a modelling research context is challenging because the iterative nature of model development conflicts with the inner logic of existing preregistration templates that presume a linear research workflow. The process of preregistration as it is currently implemented is tailored towards the hypothetico-deductive model of the scientific method. As such, the process of preregistration is completely distinct from, and precedes the implementation of the analysis plan laid out in the preregistration – analytic decisions are completely independent from the data.

In fact, preregistration has been defined as "the action of confirming an unalterable version of one's research plan prior to collecting data" (Pu et al., 2019) or prior to analysing the data (Mertens & Krypotos, 2019). In this process, researcher sequentially shifts from ideation, to study/analysis design and preregistration, to collecting and analysing the data, writing the report or manuscript, to documenting the research and publishing (Figure 1). The iterative nature of model development, however, inherently precludes this model of preregistration from being applied to model-based research in ecology and conservation.

Iterative Cycle of Model Development and Preregistration

Current preregistration practices based on NHST-focussed accounts of questionable research practices prescribe the principle of 'decision-independence', where analysis decisions are made independently of and *a priori* to analysing the data (Gelman & Loken, 2013; Srivastava, 2018). However, the modelling process may genuinely violate the principle of 'decision-independence' because it is non-linear, iterative and usually generates many interim versions of the model preceding publication. During modelling a researcher will almost always engage in state-dependent decision-making — that is, decisions about whether to alter the model in a subsequent round of model development depend on the results of model testing, evaluation and analysis at the end of the previous iteration of model development. These practices aren't necessarily questionable in and of themselves, but do have the *potential* to become questionable if they are exploited in service of artificially inflating the accuracy or precision of the model, its predictions or evaluation tests to the effect that the model is perceived to be more credible than it would be if that practice did not occur – essentially data-dependent decisions that lead to consumers of the model placing unsubstantiated belief in the reliability, validity and utility of the model and its outputs are questionable and can be mitigated through the practice of preregistration (Gould et al., 2026).

Very often in modelling, there are decisions that are necessarily dependent on the outcome of previous analytic decisions in the modelling workflow (Liu et al., 2020). Firstly, preliminary or investigatory analyses might need to be conducted before being able to specify future decision-steps in the analysis plan. For example, modellers might need to check the distribution of particular variables in order to determine how to specify the model, perform assumption checks, or check for collinearity or spatial autocorrelation. Secondly, data-dependent decisions from model checking may justifiably result in changes to either the analysis or to interpretations of the mode (Srivastava, 2018). For instance, inspection of the residuals and other model checking tests may force the modeller to return to earlier decision-points and change the planned analysis or model. Other times, the planned analysis and specified model may simply not converge, or the model is saturated and runs out of variation to apportion such that the model and or the model fitting algorithm must be respecified and re-implemented. Other times, the modelling process itself may generate knowledge and learning about the system, and the model structure changes throughout the process of model development.

Finally, sometimes there are some decisions that are too difficult to anticipate, or simply cannot be made in advance (Srivastava, 2018). This is particularly true for decisions occurring at the later phases of the modelling construction and development process, where downstream decisions depend on the outcomes of earlier decisions and outcomes of modelling analyses. Some decisions might not be able to be specified on the first attempt at writing the preregistration until the model is fully or at least close to fully specified, for example, specifying precisely what and how sensitivity analyses or uncertainty analyses will be conducted.

As illustrated above, the adaptive nature of analysis during the model development process contravenes the central principle of preregistration – decision-independence and is fundamentally incompatible with the sequential, linear process of traditional preregistration where the analysis plan is fully specified prior to observing and analysing the data.

Adaptive Preregistration

In order to address the disjuncture between the iterative nature of model development, and the single-use, deterministic preregistration template format, we take an 'expanded view of preregistration', that fits with the concept of 'adaptive preregistration' proposed by Srivastava (2018). The modeller proceeds through the model development process, shifting from ideation, preregistration, execution of the analysis and back to ideation and preregistration again. As they progress, the modeller implements parts of the analysis plan that were specified in the previous preregistration step, with the outcomes of that analysis step either informing the specification of the next analysis (Figure 4) or else resolving the preregistered decision.

This view of preregistration breaks with the current model of preregistration, wherein the author writes a single deterministic preregistration containing a rule for every decision, and sequentially shifts from ideation, preregistration to execution of the plan. Two features define adaptive preregistration:

1. It contains 'plans to deploy flexible strategies' (Srivastava, 2018) wherein the author can supply heuristic consisting of multiple different analysis or modelling strategies whose execution depends on the outcome of previous decision-points or analyses, for example, decision-trees could be preregistered prior to analysing the data (Baldwin et al., 2022).
2. The preregistration may be iterative and consist of interim preregistrations that mark phases of modelling and analysis as different parts of the data are observed (Srivastava, 2018).

Preregistering Flexibility

When preliminary analyses, model checking or other data-dependent decisions need to be made, the modeller can specify a 'flexible strategy' (Figure 2) in their preregistration by:

- stating what quantity or outcome needs to be known to move forward with the modelling and analysis,
- explaining what test or analysis will be performed to obtain this quantity or outcome, and what parts of the data will be used in this analysis, and

- describing how the results will be interpreted, listing each potential decision and its trigger, where possible.

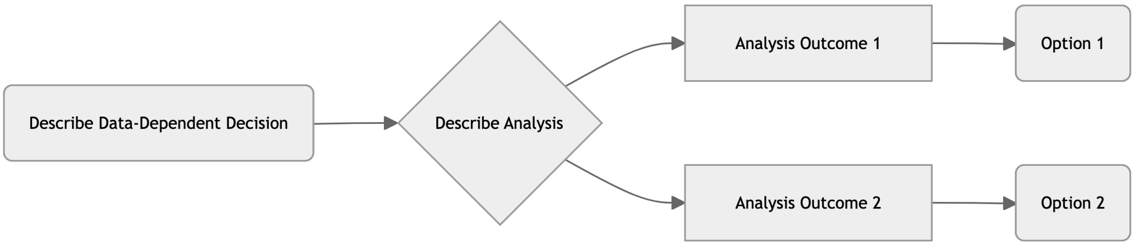

Figure 2: A general heuristic for registering flexible analyses.

Interim Preregistrations

The modeller follows an iterative process of preregistration. As they proceed through the model development process, they can shift from ideation, preregistration and execution of the analysis plan, to preregistration again. Depending on observed outcomes of the pre-specified decision-trees or registered flexible heuristics, interim preregistrations are created at multiple points in the model development process, each time there is an addition or amendment to the analysis plan.

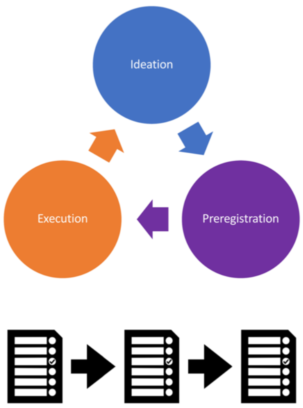

Figure 3: Adaptive preregistration can cycle through the specification, execution and ideation phases multiple times in adaptive preregistration, with interim versions of the preregistration being generated at each preregistration point.

Data-dependent decision-making and the risk of overfitting

Documenting adaptive preregistration transparently

The *Checking Problem*

Following publication of the final study, authors, reviewers, editors and readers may all engage in the process of checking a manuscript or published paper against its preregistration to verify that the study and analyses were conducted as specified in the preregistration. The process of preregistration is already cumbersome, because the dominant format of the preregistration is a single, static text-based document designed with the study authors in mind to very quickly insert the required analysis plan information, rather than to facilitate a comparison of the reported analysis and results against the planned analysis in the preregistration. In their current form, they are almost exclusively exist as "write-only media" (Pu et al., 2019). The task of model checking is further complicated by the non-linear and iterative nature of model development, and by our proposed adaptive preregistration methodology, because in most modelling studies there will likely be interim preregistrations, such that there are multiple versions of the preregistration to check against the completed analysis.

Implementing *Adaptive Preregistration* with git and GitHub

Given the iterative nature of adaptive preregistration then, the current convention of the single-use, deterministic and static text-based document format for preregistration is inadequate. Interim versions of a preregistration must be time-stamped and ideally version-controlled, such that it is clear not just that two or more versions differ from each other, and in what order they were created, but so that it is explicitly clear in how and why they differ. To facilitate preregistration checking of flexible modelling and analysis strategies, we must link the results of both final and preliminary analyses back to the specified strategy: the link between a particular triggered analysis decision, and the analysis outcome that triggered the decision must be explicit. Additionally, if the results of some preliminary analyses cause revisions to the model itself or a different modelling procedure or approach to be selected than was originally or previously planned, then the trigger for this decision must be explicitly linked to the next version of the preregistration.

We propose leveraging the features of git and GitHub to facilitate implementation both adaptive preregistration in a transparently documented manner (Figure 4), and provide a step-by-step guide.

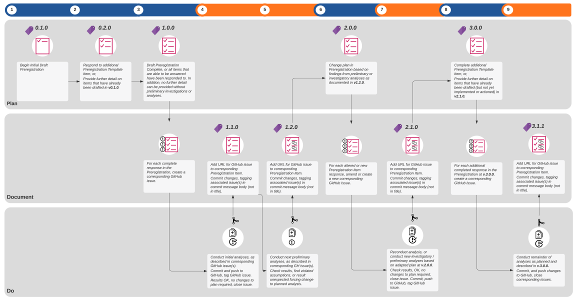

Figure 4: Adaptive Preregistration for model-based research using GitHub.

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Implementing *Adaptive Preregistration* with git and GitHub

Who is this guide for?

This guide assumes a basic understanding and knowledge of version-control tools, like git and/or GitHub. However, we have provided some background knowledge, links and references throughout this guide so that any modellers or researchers can set up and implement adaptive preregistration with git and GitHub for a new project. As a starting point, we direct readers to Braga et al. (2023), which provides an excellent introductory overview of the GitHub's functionality.

For those with basic familiarity of git and GitHub, we use a shared repository model for collaboration. If you are not yet familiar with GitHub features, such as pull requests and branches, we have linked to the relevant GitHub documentation pages where further background reading might be necessary.

We suggest some tooling that assumes use of the R statistical programming language (R Core Team (2024), see Section), since that is the most common programming language in the fields of ecology (Gao et al., 2025), however the general approach and principles described in this document is language agnostic, and does not require the use of R.

Software and Set-up Requirements

Follow the instructions at https://docs.github.com/en/get-started/git-basics/set-up-git to setup the following:

- git installation on your computer (Windows, Mac, linux supported)
- GitHub user account
- Configure authentication with GitHub from git

After you have installed and set-up the required software and GitHub account, you will need a git/GitHub repository (see Quickstart for repositories) where your preregistration document and analysis files will be version-controlled.

> Adaptive Preregistration Primer
>
> Make sure you have read the primer on Adaptive Preregistration and background before preceding with this guide.

Adaptive Preregistration using GitHub

We propose leveraging both the version-control and the project management features of GitHub (www.github.com) as a tool for implementing adaptive preregistration. GitHub is a web-interface that utilises the version-control system, 'git', to store files, track changes, and enable collaboration on computer code (Braga et al., 2023). When a document is version-controlled using Git and GitHub, each version of a document can be time-stamped, and is assigned its own unique identifier, or commit-hash[1] so that chronological records of changes to a collection of files, stored in a 'repository', are made transparent (Braga et al., 2023; Bryan & Crossman, 2008).

Why GitHub for Adaptive Preregistration?

- Because GitHub documents are 'time-stamped' the genesis of the preregistration from one version to the next is apparent.
- Moreover, using the diff view of a document on GitHub, exactly how and where a document has changed between versions is clear - additions are coloured green, deletions are coloured red, and changes within a line are highlighted.
- The project management and communication features of GitHub, such as GitHub Issues,
- tagging and release

GitHub users can collaborate by contributing, modifying and discussing existing code, reporting bugs and other problems, discover new code and data, and publish new code.

GitHub also integrates several communication features, such as GitHub Issues, GitHub Discussions, GitHub Pages, which allows users to engage in discussions, to plan and collaborate on code, and publish information to a webpage (see Box 1 in Braga et al., 2023).

The preregistration document should be saved within the project repository that contains the data and modelling and analysis code, such that any results from analyses that cause the analysis plan to change can be linked and referenced in the updated version of the plan – thus facilitating the process of model checking. By providing a method for contrasting the actual analysis undertaken against the preregistered analysis, this provides a mechanism for allowing reviewers to properly evaluate the preregistration

[1] https://docs.github.com/en/free-pro-team@latest/github/getting-started-with-github/github-glossary#commit-id

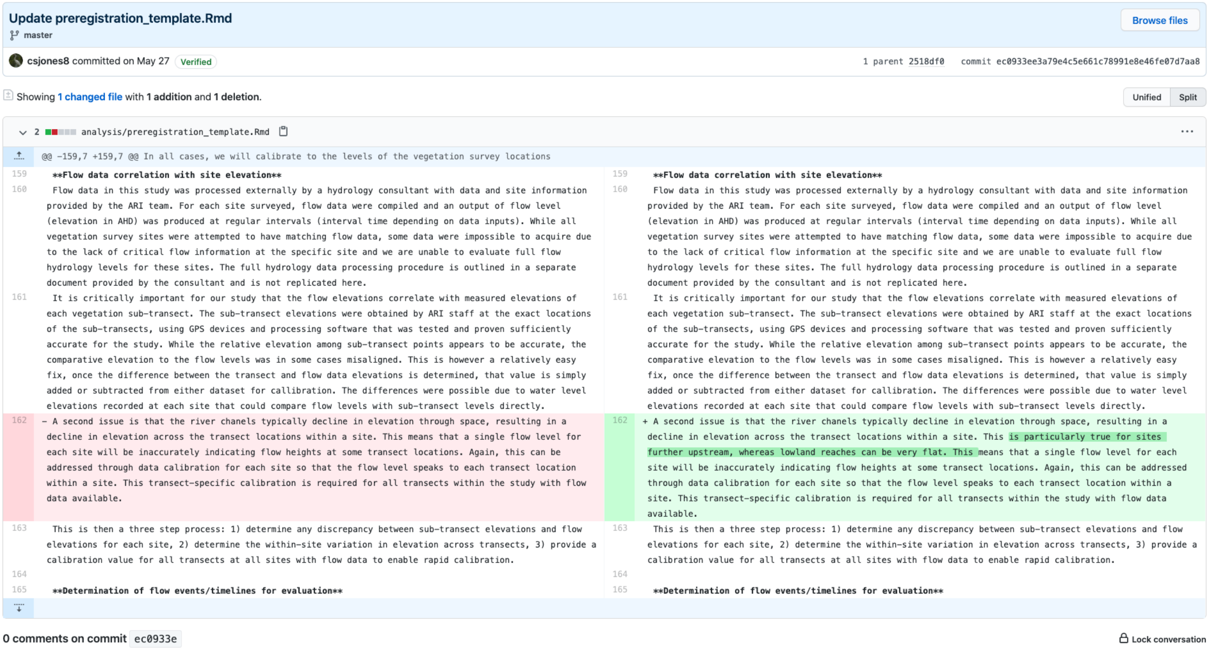


Figure 1: 'Diff View' of changes made to a preregistration in process.

against the reported analysis. It thus provides a way of marking preregistered parts of a report from non-preregistered parts of the analysis.

How it works...

GitHub is used to track amendments to the preregistration – each time an additional template item is answered, or an alteration is made to the preregistration, the change should be *committed* to the repository. *Semantic versioning* is used to document each version of the preregistration. Semantic versioning[2] is an open-source software engineering practice used to document and and communicate the development of software, in a way that allows others to depend on it (Kitzes et al., 2018). We have created our own semantic versioning scheme to be used to specifically document the incremental version of the preregistration, in tandem with GitHub's tag and release feature, each interim preregistration is tagged and released with its semantic version number. All template items that can be completed should be completed before any data analysis or modelling proceeds. The preregistration should be completed until a) no further detail can be

[2]The semantic version is a 3-part numeric identifier separated by points, e.g.: `0.0.1`, wherein the first number sequences a major version of the preregistration, the second number sequences a minor change, and the third number sequences small changes or patches, yielding: `MAJOR.MINOR.PATCH`. See https://semver.org/ for further information. Further detail on how to use semantic versioning with adaptive preregistration is summarised in Table 1.

supplied or b) no additional template item can be answered until preliminary investigations or analyses are undertaken (Figure 2, steps 1 - 3).

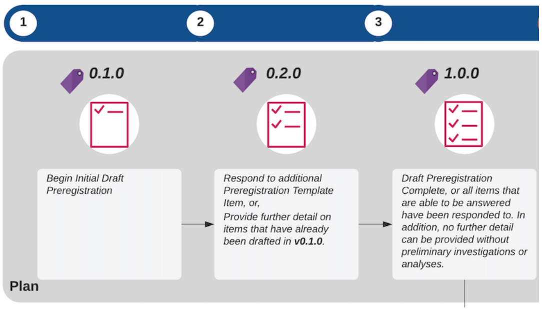


Figure 2: Initial Preregistration: All template items that can be answered should be completed prior to data analysis or modelling.

...Shifting from planning to execution...

Figure 3: Document: After the initial preregistration, document the initial preregistration using GitHub issues.

Before shifting from planning to the execution of the interim preregistration, each complete response in the interim preregistration should have a corresponding GitHub issue[3] created (Figure 3, step 3, 'Document'). Relevant discussion between analysts about how to proceed with the intended analysis or interpreting preliminary results are tracked within an issue's thread. Each issue is automatically assigned a unique number by GitHub. Any code or analysis outputs, such as figures, tables or other files, should be committed and tagged with the issue number, so that all analysis addressing that task is tracked within the issue's thread (Figure 4, step 4, 'Do'). The URL for the GitHub issue should then be added to the relevant preregistration item, the preregistration document should then be committed with the changes, and the version number of the preregistration updated in a new release.

[3] https://guides.github.com/features/issues/

... And back again.

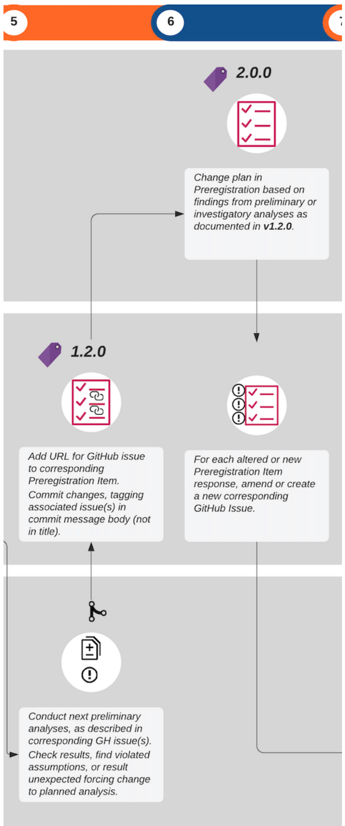


Figure 4: Adaptive Preregistration - When Plans Change

As the modelling proceeds, further detail on later phases of the analysis can be iteratively updated and preregistered. Where the results of model evaluation and analysis reveal that there are problems with the model, plans can be changed, and again the next phase can be preregistered. For example, if a researcher finds that assumptions are violated or other unexpected results occur that force a change to the planned analysis, the preregistration can be updated based on the findings of those analyses. When this occurs, the major version number of the preregistration document should be incremented (Figure 4, steps 5 - 6). Since the findings of the analyses are tracked within the relevant issue for that template item, and the issue is recorded in the preregistration itself, the trigger for the change in the plan is made explicitly clear.

For each altered or new preregistration item, the issue thread should either be updated, or a new issue created respectively (Figure 4, step 6). Those analyses are then either reconnected based on the revised plan or the new investigatory analyses are then conducted (Figure 4, step 7).

This process continues until the preregistration is complete, and the researcher can continue to execute the plan as it has been fully described in the final version of the preregistration document (Figure 4, step 8 - 9).

Flexible Strategies, Preliminary Analyses or Investigations

When a researcher needs to conduct preliminary analyses or check parts of the data before committing to a particular decision about the model or analysis, they can specify a flexible strategy in their response to that preregistration item by:

- stating what needs to be known to move forward with the modelling and analysis and why the analysis is necessary
- explaining how the researcher will test this, and what parts of the data will be used in that analysis
- describing how the results will be interpreted, listing each potential decision and the analysis result that will trigger that decision, where possible.

A step-by-step guide to implementing Adaptive Preregistration with GitHub

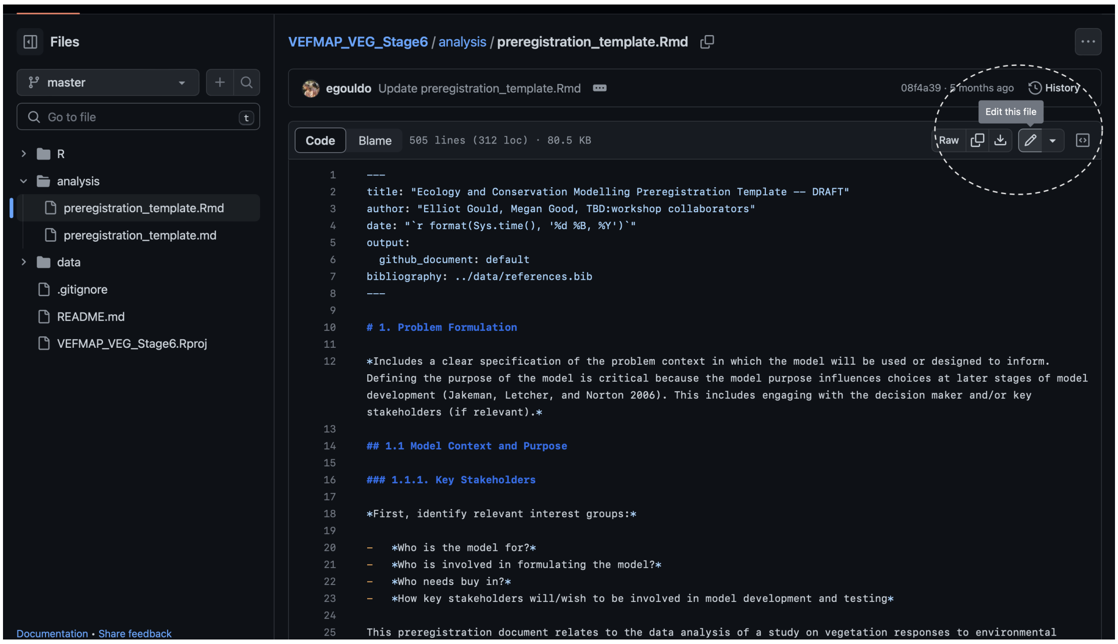
The initial preregistration phase

> Preregistration Template
>
> We have developed a template for preregistering ecological modelling studies. The template is available for use as a quarto template extension, which allows rendering to `.html`, `.pdf` or `.docx` file formats. Should you wish *not* to use quarto markdown as the format for your preregistration document, feel free to copy the relevant content into the text file format of your choice.
> Ensure you have committed a preregistration template to your project repository before proceeding with the following steps.

The first iteration of preregistration

1. Create a GitHub issue describing any tasks relevant to creation of the initial preregistration
2. Edit the document with your responses to the template items

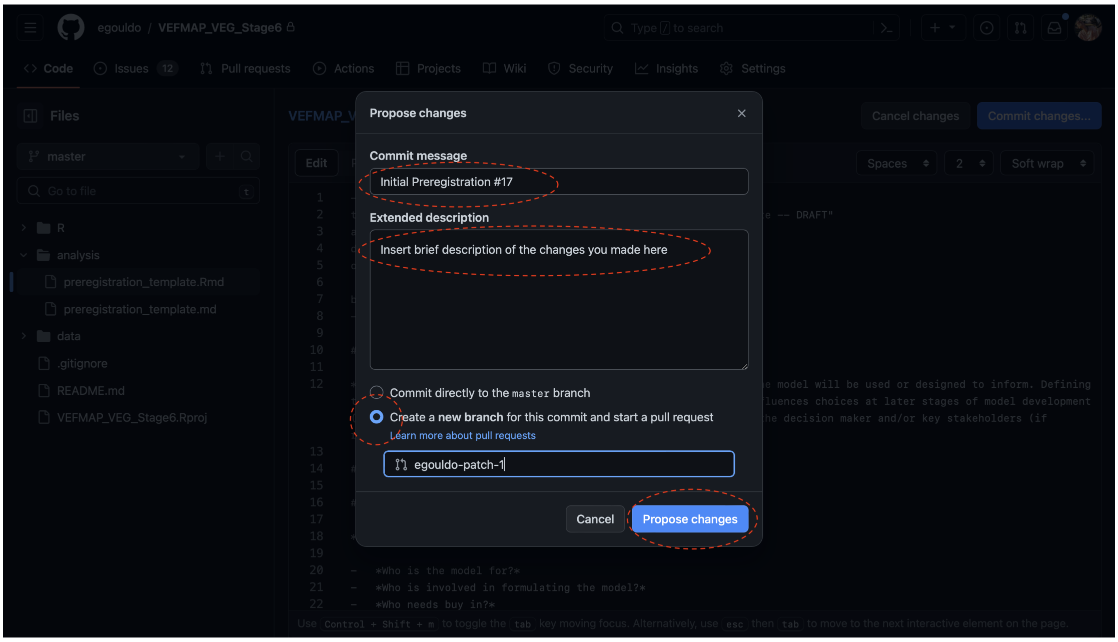
- View our preregistration template document in GitHub here.
- In your own repository containing your preregistration document, click on the 'edit document' button:



- You will be able to edit the document directly in GitHub, Edit away!

2. Commit changes

- If you are finished, or ready to save your changes by 'committing' them. There is a box called "commit changes" at the top right of the window, a pop-up window will appear, enter in the commit message box the following:
    - `Initial Preregistration #IssueNumber`. Tag the Preregistration GitHub issue number in the commit message (For this repository, it's `#17`, however if you have only created the issue for the initial preregistration, then your issue number will be `#1`).
- Fill out the extended description box briefly describing the changes you just made
- Ensure that the radio button for creating a new branch is selected, an auto-generated branch name will appear, e.g. `egouldo-patch-1`.
- Click "Propose Changes" to start a pull request:



❗ Pull Requests & the GitHub flow model of collaboration

The approach to using GitHub in this document follows the GitHub flow model for collaboration on projects. However, if you are working independently rather than collaborating on your project in a team, you can instead skip the use of pull requests and branches, and select the 'commit directly to the `default` branch' radio button above.

Note that if using the GitHub flow model demonstrated in this guide, before proceeding to the next step, you will need to make sure that the pull request is reviewed, updated if necessary, and the changes merged into the default branch once the pull request has been approved.

See the 'reviewing changes in pull requests' guide for further details.

Incrementally Updating the Preregistration Plan

Moving from general to specific: Incrementally updating the preregistration

You will likely find that some items on the template cannot be answered just yet. One reason being that the level of detail required to adequately answer the template item might not be clear yet, and more thought and time is needed, or else you need to be further along the modelling process in order to adequately answer the preregistration item. Another reason is that downstream decisions might depend on some preliminary investigations of the data – for example, you might need to explore the shape of a measured variable before you can properly specify the full model.

In the latter situation, where possible, you can preregister your preliminary analysis: describing what the preliminary analysis entails, providing a short and relevant list of plausible outcomes, and describing for each of those possible outcomes how the dependent or down-stream decision-point will change depending on the outcome.

Using GitHub to update the plan

We will use GitHub to track amendments to the preregistration for the two situations described above (a. adding more detail, answering incomplete questions; b. Responding to findings of a preliminary analysis already described in your preregistration).

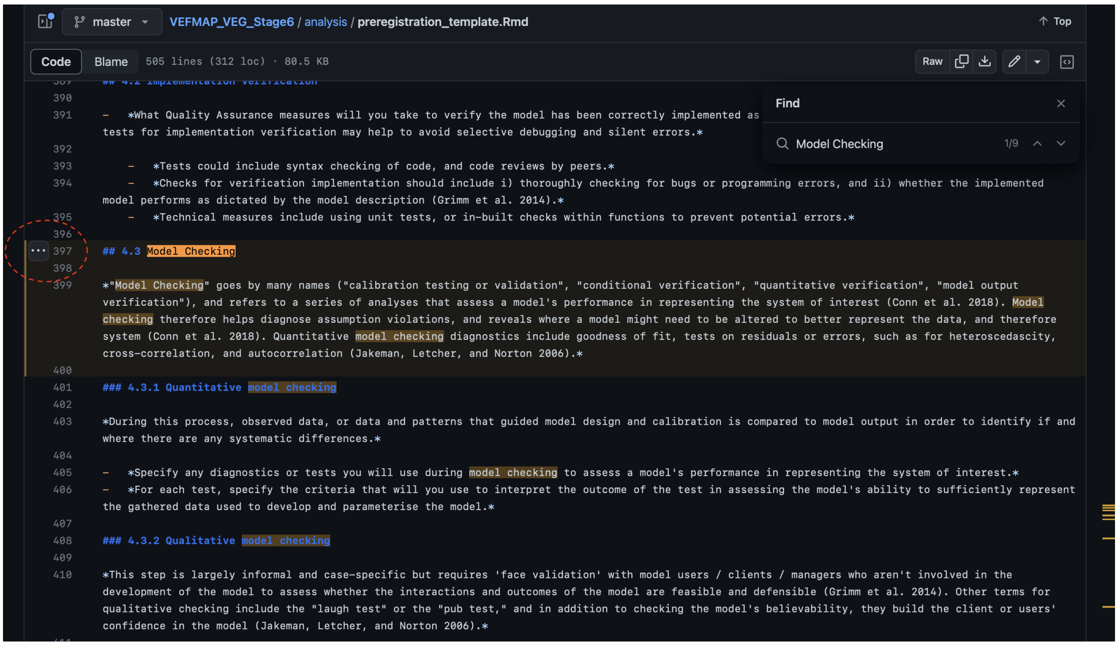
To make the changes to the preregistration document, follow the same instructions as for creating the initial preregistration above - simply edit the preregistration in GitHub, and commit the changes.

As for the situation where you are responding to findings of a preliminary analysis that you have already described in your preregistration, see the below sections on `Moving from planning to doing` and `Shifting back to planning from doing`.

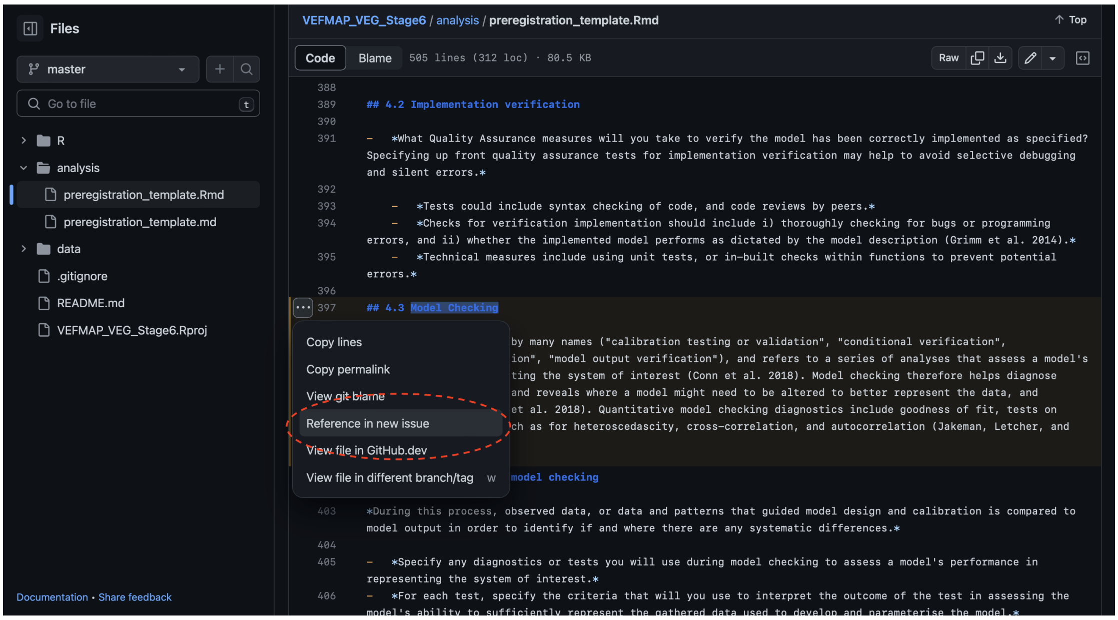
Moving from planning to doing

Before you begin to do or start anything you've planned within your preregistration, you **must** create a GitHub issue for that item, phase, or step in the modelling process, follow these steps:

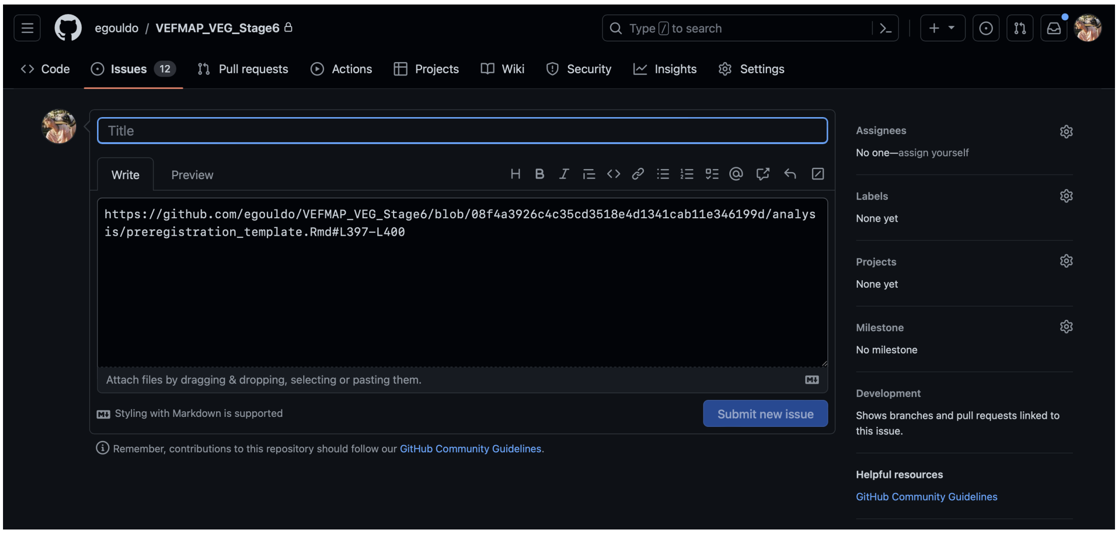
1. View the preregistration document in GitHub, and click on the line-numbers on the left to highlight the preregistration template item text, as well as your response to the item. For example, let's say you want to start working the model checking, you would highlight the following in the document. Note that once you've highlighted some lines, a little box with an ellipsis appears on top of the lines numbers.



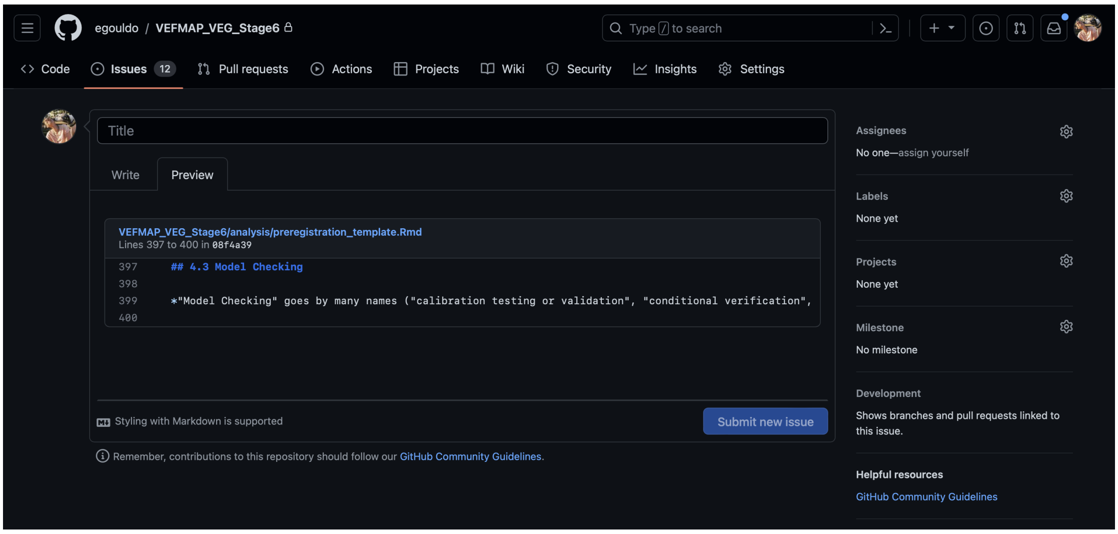
2. Click on the box with the ellipsis, and a pop-up dialogue box will appear with 4 options: `Copy lines`, `Copy permalink`, `View git blame`, `References in new issue`. Select the last option `Reference in new issue`.



3. This will open up a new window with a new GitHub Issue for you to complete. Note that part of the issue description has already been filled out for you, this is a reference to the lines you just highlighted:



Which is cool, because if you open the `Preview` tab to view the preview of your issue, it will look like this:



Add any further information or technical details about going forward and working on this task, as necessary. Submit the issue to save. Any work, and subsequent commits on this topic should be tagged with the issue number. We now have a mechanism for linking your actual analysis and actions to the planned analysis in the preregistration document.

Dealing with unexpected outcomes, and changing plans: Shifting back to planning from doing

Sometimes assumptions are violated, or there are other unexpected outcomes of preliminary analyses, and you will need to change plans. How do we deal with this?

- Add the URL to the relevant GitHub issue under the corresponding Preregistration Item.
- Add any the URL for any linked commits to the Preregistration Item.
- Revise the Preregistration, where necessary by editing the Preregistration document in GitHub, as per previous instructions.
- Commit the changes with the following commit message header (or similar): `Preregistration Deviation #issue`.
- Describe the trigger for the Preregistration revisions in brief detail within the commit message body, adding the link to the relevant GitHub issue containing the tagged commits that triggered the revision.

'Living preregistration' - tracking changes to the Preregistration Draft

We will leverage GitHub's tagging and release feature in conjunction with Semantic Versioning to document and track changes to the preregistration document.

GitHub releases mark software iterations that you can release and deploy[4]. Within the context of adaptive preregistration, we can use releases to mark interim versions of the preregistration, as well as to take a snapshot of the state of the entire repository at that moment in time.

A summary of the types of preregistration changes, and the corresponding required version and GitHub features are presented in Table 1. Note that anything up until the complete preregistration should be additionally tagged as a 'pre-release' to communicate that the initial preregistration is not yet complete.

Using the `fledge::` R package for semantic versioning and to document preregistration changes

The package `fledge::` can be used right from the R console to bump the version number, make releases, and auto catalogue a chronological history of changes recorded a `NEWS.md` file that is updated and published with every release. A detailed guide to using `fledge::` can be found here https://fledge.cynkra.com/articles/fledge.html.

[4]Instructions for creating releases are here: https://help.github.com/en/github/administering-a-repository/managing-releases-in-a-repository

This package is not a mandatory component of the process, however can be a useful tool for implementing semantic versioning, tagging and releasing. All of these actions can be implemented in the GitHub browser with a point and click interface.

Table 1: Types of changes to the preregistration and corresponding actions and versioning adjustments

Preregistration Change or Action	GitHub Mechanism	Semantic Versioning
Changes preceding initial completion of preregistration	Tag and Mark as 'pre-release'	`Minor` increment. E.g. `0.2.0` The `Major` value should not exceed `0`. `fledge::bump_-version("minor")`
Incremental Change, e.g. respond to new question.	Tag and Release	`Minor` increment. E.g. `0.2.0` The `Major` value should not exceed `0`. `fledge::bump_-version("minor")`
Initial Preregistration complete. Note that 'complete' needs to be decided. Is complete after all preliminary and investigatory analyses have been done and the final Preregistration is complete, or is it the first iteration, where no more items can be filled out until some initial analyses are undertaken?	Tag and Release	`Major` increment to `1.0.0`. `fledge::bump_-version("major");` `fledge::finalize_-version()`.
Major change to a complete Preregistration – no looking or analysing at data.	Tag and Release	`Major` increment `>1.0.0`. `fledge::bump_-version("major");` `fledge::finalize_-version()`.

Preregistration Change or Action	GitHub Mechanism	Semantic Versioning
Linking implemented work to the Preregistration Item	Update Preregistration with GH Issue URL and all relevant commit hashes pertaining to that Preregistration Item + Commit. Tag and Release that preregistration commit.	`Minor` increment. `fledge::bump_version("minor")`.
Data-dependent changes with registered flexibility Adapting the Preregistration on the basis of results from preliminary or investigatory analyses	Update Preregistration with GH Issue Link and all relevant commit hashes pertaining to that Preregistration Item + Commit. Respond to dependent incomplete Preregistration Items or Update existing / already completed Preregistration Items + Commit. Tag and Release.	`Major` increment. `fledge::bump_version("major")`.
Minor mistakes or changes	Changes including small typos, or minor elaborations on existing specified plans	`Patch` increment. E.g. `0.1.2`. `fledge::bump_version("patch")`

References

Braga, P. H. P., Hébert, K., Hudgins, E. J., Scott, E. R., Edwards, B. P. M., Sánchez Reyes, L. L., Grainger, M. J., Foroughirad, V., Hillemann, F., Binley, A. D., Brookson, C. B., Gaynor, K. M., Shafiei Sabet, S., Güncan, A., Weierbach, H., Gomes, D. G. E., & Crystal-Ornelas, R. (2023). Not just for programmers: How GitHub can accelerate collaborative and reproducible research in ecology and evolution. *Methods in Ecology and Evolution*, *14*(6), 1364–1380. https://doi.org/10.1111/2041-210X.14108

Bryan, B. A., & Crossman, N. D. (2008). Systematic regional planning for multiple objective natural resource management. *Journal of Environmental Management*, *88*(4), 1175–1189. https://doi.org/10.1016/j.jenvman.2007.06.003

Gao, M., Ye, Y., Zheng, Y., & Lai, J. (2025). A comprehensive analysis of r's application in ecological research from 2008 to 2023. *Journal of Plant Ecology*, *18*(1), rtaf010. https://doi.org/10.1093/jpe/rtaf010

Kitzes, J., Turek, D., & Deniz, F. (2018). *The practice of reproducible research: Case studies and lessons from the data-intensive sciences.* (J. Kitzes, D. Turek, & F. Deniz, Eds.; pp. 1–319). University of California Press. http://www.practicereproducibleresearch.org

R Core Team. (2024). *R: A language and environment for statistical computing.* R Foundation for Statistical Computing. https://www.R-project.org/

Part III: Preregistration Template for Ecological Modelling

Adaptive Preregistration for Modelling

A template suitable for use in Ecology and Conservation with accompanying rationale and explanation.

Problem Formulation

> **i Rationale & Explanation**
>
> This section specifies the decision-making context in which the model will be used or the intended scope and context of conclusions. Important components include the decision maker and stakeholders (including experts) and their view on: i) the nature of the problem or decision addressed and how the scope of the modelling tool fits within the (broader) context (i.e. model purpose; ii) the spatial and temporal scales relevant to the decision context; iii) specified desired outputs; iv) role and inclusion in model development and testing; v) whether they foresee unacceptable outcomes that need to be represented in the model (i.e. as constraints), and; vi) what future scenarios does the model need to account for (noting this may be revised later). It should also provide a summary of the domain of applicability of the model, and reasonable extrapolation limits (Grimm et al., 2014).

Model Context and Purpose

> **i Rationale & Explanation**
>
> Defining the purpose of the model is critical because the model purpose influences choices at later stages of model development (Jakeman et al., 2006). Common model purposes in ecology include: gaining a better qualitative understanding of the target system, synthesising and reviewing knowledge, and providing guidance for management and decision-making (Jakeman et al., 2006). Note that modelling objectives are distinct from the analytical objectives of the model.

> The scope of the model includes temporal and spatial resolutions, which should also be defined here (Mahmoud et al., 2009). Any external limitations on model development, analysis and flexibility should also be outlined in this section (Jakeman et al., 2006).

Key stakeholders and model users

> Preregistration Item
>
> *Identify relevant interest groups:*
>
> - ☐ Who is the model for?
> - ☐ Who is involved in formulating the model?
> - ☐ How will key stakeholders be involved in model development?
> - ☐ Describe the decision-making context in which the model will be used (if relevant).

Model purpose, context and problem context

> Preregistration Item
>
> *Briefly outline:*
>
> - ☐ the ecological problem,
> - ☐ the decision problem (if relevant), including the decision-trigger and any regulatory frameworks relevant to the problem,
> - ☐ how the model will address the problem, being clear about the scope of the model i.e. is the model addressing the whole problem, or part of it? Are there any linked problems that your model should consider?
> - ☐ Ensure that you specify any focal taxa and study objectives.

Analytical objectives

Explanation

How will the model be analysed, what analytical questions will the model be used to answer? For example, you might be using your model in a scenario analysis to determine which management decision is associated with minimum regret or the highest likelihood of improvement. Other examples from ecological decision-making include: to compare the performance of alternative management actions under budget constraint (Fraser et al., 2017) to search for robust decisions under uncertainty (McDonald-Madden et al., 2008), to choose the conservation policy that minimises uncertainty [insert ref]. See other examples in (Moallemi et al., 2019).

Preregistration Item

Provide detail on the analytical purpose and scope of the model:

- ☐ How will the model be analysed and what analytical questions will the model be used to answer?
- ☐ Candidate decisions should be investigated and are specified a priori. Depending on the modelling context, they may be specified by stakeholders, model users or the analyst (Moallemi et al., 2019).
  - ☐ Describe the method used to identify relevant management actions and
  - ☐ specify management actions to be considered included in the model.
  - ☐ Are there potentially unacceptable management or policy outcomes identified by stakeholders that should be captured in the model, i.e. as constraints?
- ☐ Are there scenarios that model inputs or outputs that must accommodated? Scenarios should be set a priori, (i.e. before the model is built, Moallemi et al., 2019) and may be stakeholder-defined or driven by the judgement of the modeller or other experts (Mahmoud et al., 2009).
  - ☐ If relevant, describe what processes you will use to elicit and identify relevant scenarios, e.g. literature review, structured workshops with stakeholders or decision-makers.
  - ☐ Specify scenarios under which decisions are investigated.

Logistical Constraints

> Preregistration Item
>
> - ☐ What degree of flexibility is required from the model? Might the model need to be quickly reconfigured to explore new scenarios or problems proposed by clients / managers / model-users?
> - ☐ Are there any limitations on model development analysis and flexibility, such as time or budget constraints, for example, does a model need to be deployed rapidly?
>   - ☐ When must the model be completed by, e.g. to help make a decision?

Model Scope, Scale and Resolution

> Preregistration Item
>
> - ☐ The choice of a model's boundaries is closely linked to the choice of how finely to aggregate the behaviour within the model (Jakeman et al., 2006) - what is the intended scale, and resolution of the model (temporal, spatial or otherwise)?
> - ☐ Where is the boundary of the modelled system? Everything outside beyond the boundary and not crossing it is to be ignored within the domain of the model, and everything crossing the boundary is to be treated as external forcing (known/unknown), or else as model outputs (observed, or not, Jakeman et al., 2006).

Intended application of results

> Explanation
>
> Preregistration Items in this section are relevant to model transferability (Yates et al., 2018) and constraints on generality in model analysis interpretation. How far do can the results be extrapolated based on the study design (data + model + analysis)? For instance, if there are many confounding variables and not enough spatial / environmental replication, then making broader more general claims beyond the stated boundaries of the model (Section) may not be warranted. However, larger generalisations about results may be acceptable if the data comes from experimentally manipulated or controlled systems.

Preregistration Item

- ☐ What is the intended domain in which the model is to be applied? Are there any reasonable extrapolation limits beyond which you expect the model should not be applied (Grimm et al., 2014)?

Scenario Analysis Operationalisation

Preregistration Item (delete as necessary)

- ☐ How will you operationalise any scenarios identified in 1.1.3? For example, how will you operationalise any qualitative changes of interest, such as , ‘deterioration’ or ‘improvement’?
- ☐ Describe how you will evaluate and distinguish the performance of alternative scenario outcomes
- ☐ Justify or otherwise explain how you chose these measures and determined performance criteria in relation to the analytical objectives, model purpose and modelling context, such as the risk attitudes of decision-makers and stakeholders within this system

Define Conceptual Model

Explanation

Conceptual models underpin the formal or quantitative model (Cartwright et al., 2016). The conceptual model describes the biological mechanisms relevant to the ecological problem and should capture basic premises about how the target system works, including any prior knowledge and assumptions about system processes. Conceptual models may be represented in a variety of formats, such as influence diagrams, linguistic model block diagram or bond graphs, and these illustrate how model drivers are linked to both outputs or observed responses, and internal (state) variables (Jakeman et al., 2006).

Choose elicitation and representation method

Preregistration Item

- ☐ Describe what method you will use to elicit or identify the conceptual model. Some common methods include interviews, drawings, and mapping techniques including influence diagrams, cognitive maps and Bayesian belief networks (Moon et al., 2019). It is difficult to decide and justify which method is most appropriate, see Moon et al. (2019) for guidance addressing this methodological question.
- ☐ Finally, how do you intend on representing the final conceptual model? This will likely depend on the method chosen to elicit the conceptual model.

Explain Critical Conceptual Design Decisions

Preregistration Item

List and explain critical conceptual design decisions (Grimm et al., 2014), including:

- ☐ spatial and temporal scales,
- ☐ selection of entities and processes,
- ☐ representation of stochasticity and heterogeneity,
- ☐ consideration of local versus global interactions, environmental drivers, etc.
- ☐ Explain and justify the influence of particular theories, concepts, or earlier models against alternative conceptual design decisions that might lead to alternative model structures.

Model assumptions and uncertainties

Preregistration Item

Specify key assumptions and uncertainties underlying the model design, describing how uncertainty and variation will be represented in the model (Moallemi et al., 2019). Sources of uncertainty may include:

- ☐ exogenous uncertainties affecting the system,
- ☐ parametric uncertainty in input data and
- ☐ structural / conceptual nonparametric uncertainty in the model.

Identify predictor and response variables

Explanation

The identification and definition of primary model input variables should be driven by scenario definitions, and by the scope of the model described in the problem formulation phase (Mahmoud et al., 2009).

Preregistration Item

Identify and define system system variables and structures, referencing scenario definitions, and the scope of the model as described within problem formulation:

- ☐ What variables would support taking this action or making this decision?
- ☐ What additional variables may interact with this system (things we can't control, but can hopefully measure)?
- ☐ What variables have not been measured, but may interact with the system (often occurs in field or observational studies)?
- ☐ What variables are index or surrogate measures of variables that we cannot or have not measured?
- ☐ In what ways do we expect these variables to interact (model structures)?
- ☐ Explain how any key concepts or terms within problem or decision-making contexts, such as regulatory terms, will be operationalised and defined in a biologically meaningful way to answer the research question appropriately?

Define prior knowledge, data specification and evaluation

Explanation

This section specifies the plan for collecting, processing and preparing data available for parameterisation, determining model structure, and for scenario analysis. It also allows the researchers to disclose any prior interaction with the data.

Collate available data sources that could be used to parameterise or structure the model

Preregistration Item

For pre-existing data (delete as appropriate):

- ☐ Document the identity, quantity and provenance of any data that will be used to develop, identify and test the model.
- ☐ For each dataset, is the data open or publicly available?
- ☐ How can the data be accessed? Provide a link or contact as appropriate, indicating any restrictions on the use of data.
- ☐ Date of download, access, or expected timing offuture access.
- ☐ Describe the source of the data - what entity originally collected this data? (National Data Set, Private Organisational Data, Own Lab Collection, Other Lab Collection, External Contractor, Meta-Analysis, Expert Elicitation, Other).
- ☐ Codebook and meta-data. If a codebook or other meta-data is available, link to it here and / or upload the document(s).
- ☐ Prior work based on this dataset - Have you published / presented any previous work based on this dataset? Include any publications, conference presentations (papers, posters), or working papers (in-prep, unpublished, preprints) based on this dataset you have worked on.
- ☐ Unpublished Prior Research Activity - Describe any prior but unpublished research activity using these data. Be specific and transparent.
- ☐ Prior knowledge of the current dataset - Describe any prior knowledge of or interaction with the dataset before commencing this study. For example, have you read any reports or publications about this data?
- ☐ Describe how the data is arranged, in terms of replicates and covariates.

Sampling Plan (for data you will collect, delete as appropriate):

- ☐ Data collection procedures - Please describe your data collection process, including how sites and transects or any other physical unit were selected and arranged. Describe any inclusion or exclusion rules, and the study timeline.
- ☐ Sample Size - Describe the sample size of your study.
- ☐ Sample Size Rationale - Describe how you determined the appropriate sample size for your study. It could include feasibility constraints, such as time, money or personnel.
- ☐ If sample size cannot be specified, specify a stopping rule - i.e. how will you decide when to terminate your data collection?

Data Processing and Preparation

Preregistration Item

- ☐ Describe any data preparation and processing steps, including manipulation of environmental layers (e.g. standardisation and geographic projection) or variable construction (e.g. Principal Component Analysis).

Describe any data exploration or preliminary data analyses.

Explanation

In most modelling cases, it is necessary to perform preliminary analyses to understand the data and check that assumptions and requirements of the chosen modelling procedures are met. Data exploration prior to model fitting or development may include exploratory analyses to check for collinearity, spatial and temporal coverage, quality and resolution, outliers, or the need for transformations (Yates et al., 2018).

Preregistration Item

For each separate preliminary or investigatory analysis:

- ☐ State what needs to be known to proceed with further decision-making about the modelling procedure, and why the analysis is necessary.
- ☐ Explain how you will implement this analysis, as well as any techniques you will use to summarise and explore your data.
- ☐ What method will you use to represent this analysis (graphical, tabular, or otherwise, describe)
- ☐ Specify exactly which parts of the data will be used
- ☐ Describe how the results will be interpreted, listing each potential analytic decision, as well as the analysis finding that will trigger each decision, where possible.

Data evaluation, exclusion and missing data

Explanation

Documenting issues with reliability is important because data quality and ecological relevance might be constrained by measurement error, inappropriate experimental design, and heterogeneity and variability inherent in ecological systems (Grimm et al., 2014). Ideally, model input data should be internally consistent across temporal and spatial scales and resolutions, and appropriate to the problem at hand (Mahmoud et al., 2009).

Preregistration Item

- ☐ Describe how you will determine how reliable the data is for the given model purpose. Ideally, model input data should be internally consistent across temporal and spatial scales and resolutions, and appropriate to the problem at hand
- ☐ Document any issues with data reliability.
- ☐ How will you determine what data, if any, will be excluded from your analyses?
- ☐ How will outliers be handled? Describe rules for identifying outlier data, and for excluding a site, transect, quadrat, year or season, species, trait, etc.
- ☐ How will you identify and deal with incomplete or missing data?

Conceptual model evaluation

Preregistration Item

- ☐ Describe how your conceptual model will be critically evaluated. Evaluation includes both the completeness and suitability of the overall model structure.
- ☐ How will you critically assess any simplifying assumptions (Augusiak et al., 2014)?
- ☐ Will this process will include consultation or feedback from a client, manager, or model user.

Formalise and Specify Model

Explanation

In this section describe what quantitative methods you will use to build the model/s, explain how they are relevant to the client/manager/user's purpose.

Model class, modelling framework and approach

Explanation

Modelling approaches can be described as occurring on a spectrum from correlative or phenomenological to mechanistic or process-based (Yates et al., 2018); where correlative models use mathematical functions fitted to data to describe underlying processes, and mechanistic models explicitly represent processes and details of component parts of a biological system that are expected to give rise to the data (White & Marshall, 2019). A model 'class,' 'family' ' or 'type' is often used to describe a set of models each of which has a distinct but related sampling distribution (C. C. Liu & Aitkin, 2008). The model family is driven by choices about the types of variables covered and the nature of their treatment, as well as structural features of the model, such as link functions, spatial and temporal scales of processes and their interactions (Jakeman et al., 2006).

Preregistration Item

- [ ] Describe what modelling framework, approach or class of model you will use to implement your model and relate your choice to the model purpose and analytical objectives described in 1.1.2 and 1.1.3.

Choose model features and family

Explanation

All modelling approaches require the selection of model features, which conform with the conceptual model and data specified in previous steps (Jakeman et al., 2006). The choice of model are determined in conjunction with features are selected. Model features include elements such as the functional form of interactions, data structures, measures used to specify links, any bins or discretisation of continuous variables. It is usually difficult to change fundamental features of a model beyond an early stage of model development, so careful thought and planning here is useful to the modeller (Jakeman et al., 2006). However, if changes to these fundamental aspects of the model do need to change, document how and why these choices were made, including any results used to support any changes in the model.

Operationalising Model Variables

Preregistration Item

- [ ] For each response, predictor, and covariate, *specify how these variables will be operationalised in the model. This should relate directly to the analytical and/or management objectives specified during the problem formulation phase. Operationalisations could include: the extent of a response, an extreme value, a trend, a long-term mean, a probability distribution, a spatial pattern, a time-series, qualitative change, such as a direction of change or, the frequency, location, or probability of some event occuring. Specify any treatment of model variables, including whether they are lumped / distributed, lienar / non-linear, stochastic / deterministic (Jakeman et al., 2006)* .
- [ ] *Provide a rationale for your choices, including why plausible alternatives under consideration were not chosen, and relate your justification bacj to the purpose, objectives, prior knowledge and or logistical constraints specified in the problem formulation phase (Jakeman et al., 2006).*

Choose model family

Preregistration Item

- [ ] Specify which family of statistical distributions you will use in your model, and describe any transformations, or link functions.
- [ ] Include in your rational for selection, detail about which variables the model outputs are likely sensitive to, what aspects of their behaviour are important, and any associated spatial or temporal dimensions in sampling.

Describe *approach* for identifying model structure

Explanation

This section relates to the process of determining the best/most efficient/parsimonious representation of the system at the appropriate scale of concern (Jakeman et al., 2006) that best meets the analytical objectives specified in the problem formulation phase. Model structure refers to the choice of variables included in the model, and the nature of the relationship among those variables. Approaches to finding model structure and parameters may be knowledge-supported, or data-driven (Boets et al., 2015). Model selection methods can include traditional inferential approaches such as unconstrained searches of a dataset for patterns that explain variations in the response variable, or use of ensemble-modelling methods (Barnard et al., 2019). Ensemble modelling procedures might aim to derive a single model, or a multi-model average (Yates et al., 2018). Refining actions to develop a model could include iteratively dropping parameters or adding them, or aggregating / disaggregating system descriptors, such as dimensionality and processes (Jakeman et al., 2006).

Preregistration Item

- ☐ Specify what approach and methods you will use to identify model structure and parameters.
- ☐ If using a knowledge-supported approach to deriving model structure (either in whole or in part), specify model structural features, including:
    - the functional form of interactions (if any)
    - data structures,
    - measures used to specify links,
    - any bins or discretisation of continuous variables (Jakeman et al., 2006),
    - any other relevant features of the model structure.

Describe parameter estimation technique and performance criteria

Explanation

Before calibrating the model to the data, the performance criteria for judging the calibration (or model fit) are specified. These criteria and their underlying assumptions should reflect the desired properties of the parameter estimates / structure (Jakeman et al., 2006). For example, modellers might seek parameter estimates that are robust to outliers, unbiased, and yield appropriate predictive performance. Modellers will need to consider whether the assumptions of the estimation technique yielding those desired properties are suited to the problem at hand. For integrated or sub-divided models, other considerations might include choices about where to disaggregate the model for parameter estimation; e.g. spatial sectioning (streams into reaches) and temporal sectioning (piece-wise linear models) (Jakeman et al., 2006).

Parameter estimation technique

Preregistration Item

- ☐ Specify what technique you will use to estimate parameter values, and how you will supply non-parametric variables and/or data (e.g. distributed boundary conditions). For example, will you calibrate all variables simultaneously by optimising fit of model outputs to observations, or will you parameterise the model in a piecemeal fashion by either direct measurement, inference from secondary data, or some combination (Jakeman et al., 2006).
- ☐ Identify which variables will be parameterised directly, such as by expert elicitation or prior knowledge.
- ☐ Specify which algorithm(s) you will use for any data-driven parameter estimation, including supervised, or unsupervised machine learning, decision-tree, K-nearest neighbour or cluster algorithms (Z. Liu et al., 2018).

Parameter estimation / model fit performance criteria

Preregistration Item

- ☐ Specify which suite of performance criteria you will use to judge the performance of the model. Examples include correlation scores, coefficient of determination, specificity, sensitivity, AUC, etcetera (Yates et al., 2018).
- ☐ Relate any underlying assumptions of each criterion to the desired properties of the model, and justify the choice of performance metric in relation
- ☐ Explain how you will identify which model features or components are significant or meaningful.

Model assumptions and uncertainties

Preregistration Item

- ☐ Specify assumptions and key uncertainties in the formal model. Describe what gaps exist between the model conception, and the real-world problem, what biases might this introduce and how might this impact any interpretation of the model outputs, and what implications are there for evaluating model-output to inform inferences or decisions?

Specify formal model(s)

Explanation

Once critical decisions have been made about the modelling approach and method of model specification, the conceptual model is translated into the quantitative model.

Preregistration Item

- ☐ Specify all formal models
  - ☐ Note, For data-driven approaches to determining model structure and or parameterisation, it may not be possible to respond to this preregistration item. In such cases, explain why this is the case, and how you will document the model(s) used in the final analysis.
- ☐ For quantitative model selection approaches, including ensemble modelling, specify each model used in the candidate set, including any null or full/global model.

Model Calibration, Validation & Checking

Model calibration and validation scheme

Explanation

This section pertains to any data calibration, validation or testing schemes that will be implemented. For example, the model may be tested on data independent of those used to parameterise the model (external validation), or the model may be cross-validated on random sub-samples of the data used to parameterise the model (internal cross-validation) (Barnard et al., 2019; Yates et al., 2018). For some types of models, hyper-parameters are estimated from data, and may be tuned on further independent holdouts of the training data, ("validation data").

Preregistration Item

- ☐ Describe any data calibration, validation and testing scheme you will implement, including any procedures for tuning or estimating model hyper-parameters (if any).

Describe calibration/validation data

Explanation & Rationale

The following items pertain to properties of the *datasets* used for calibration (training), validation, and testing.

Preregistration Item

If partitioning data for cross-validation or similar approach (delete as needed):

- ☐ Describe the approach specifying the number of folds that will be created, the relative size of each fold, and any stratification methods used for ensuring evenness of groups between folds and between calibration / validation data?

If using external / independent holdout data for model testing and evaluation (delete as needed):

- ☐ Which data will be used as a the testing data? What method will you be used for generating training / test data subsets?

- ☐ Describe any known differences between the training/validation and testing datasets, the relative size of each, as well as any stratification methods used for ensuring evenness of groups between data sets?
- ☐ It is preferable that any independent data used for model testing remains unknown to modellers during the process of model development, please describe the relationship modellers have to model validation data, will independent datasets be known or accessible to any modeller or analyst?

Implementation verification

Explanation & Examples

Model implementation verification is the process of ensuring that the model has been correctly implemented, and that the model performs as described by the model description (Grimm et al., 2014). This process is distinct from model checking, which assesses the model's performance in representing the system of interest (Conn et al., 2018).

- Checks for verification implementation should include i) thoroughly checking for bugs or programming errors, and ii) whether the implemented model performs as described by the model description (Grimm et al., 2014).
- Qualitative tests could include syntax checking of code, and peer-code review (Ivimey et al., 2023). Technical measures include using unit tests, or in-built checks within functions to prevent potential errors.

Preregistration Item

- ☐ What Quality Assurance measures will you take to verify the model has been correctly implemented? Specifying a priori quality assurance tests for implementation verification may help to avoid selective debugging and silent errors.

Model checking

Rationale & Explanation

"Model Checking" goes by many names ("conditional verification", "quantitative verification", "model output verification"), and refers to a series of analyses that assess a model's performance in representing the system of interest (Conn et al., 2018). Model checking aids in diagnosing assumption violations, and reveals where a model might need to be altered to better represent the data, and therefore system (Conn et al., 2018). Quantitative model checking diagnostics include goodness of fit, tests on residuals or errors, such as for heteroscedascity, cross-correlation, and autocorrelation (Jakeman et al., 2006).

Quantitative model checking

Preregistration Item

During this process, observed data, or data and patterns that guided model design and calibration, are compared to model output in order to identify if and where there are any systematic differences.

- ☐ Specify any diagnostics or tests you will use during model checking to assess a model's performance in representing the system of interest.
- ☐ For each test, specify the criteria that will you use to interpret the outcome of the test in assessing the model's ability to sufficiently represent the gathered data used to develop and parameterise the model.

Qualitative model checking

Explanation

This step is largely informal and case-specific, but requires, 'face validation' with model users / clients / managers who aren't involved in the development of the model to assess whether the interactions and outcomes of the model are feasible an defensible (Grimm et al., 2014). This process is sometimes called a ,"laugh test" or a "pub test" and in addition to checking the model's believability, it builds the client's confidence in the model (Jakeman et al., 2006). Face validation could include structured walk-throughs, or presenting descriptions, visualisations or summaries of model results to experts for assessment.

Preregistration Item

- ☐ Briefly explain how you will qualitatively check the model, and whether and how you will include users and clients in the process.

Assumption Violation Checks

Preregistration Item

The consequences of assumption violations on the interpretation of results should be assessed (Araújo et al., 2019).

- ☐ Explain how you will demonstrate robustness to model assumptions and check for violations of model assumptions.
- ☐ If you cannot perform quantitative assumption checks, describe what theoretical justifications would justify a lack of violation of or robustness to model assumptions.
- ☐ If you cannot demonstrate or theoretically justify violation or robustness to assumptions, explain why not, and specify whether you will discuss assumption violations and their consequences for interpretation of model outputs.
- ☐ If assumption violations cannot be avoided, explain how you will explore the consequences of assumption violations on the interpretation of results (To be completed in interim iterations of the preregistration, only if there are departures from assumptions as demonstrated in the planned tests above).

Model Validation and Evaluation

Explanation

The model validation & evaluation phase comprises a suite of analyses that collectively inform inferences about whether, and under what conditions, a model is suitable to meet its intended purpose (Augusiak et al., 2014). Errors in design and implementation of the model and their implication on the model output are assessed. Ideally independent data is used against the model outputs to assess whether the model output behaviour exhibits the required accuracy for the model's intended purpose. The outcomes of these analyses build confidence in the model applications and increase understanding of model strengths and limitations. Model evaluation including, model analysis, should complement model checking. It should evaluate model checking, and consider over-fitting and extrapolation. The higher the proportion of calibrated, or uncertain parameters, "the greater the risk that the model seems to work correctly, but for the wrong reasons" (citaiton). Evaluation thus complements model checking because we can rule out the chance that the model fits the calibration data well, but has not captured the relevant ecological mechanisms of the system pertinent to the research question or the decision problem underpinning the model (Grimm et al., 2014). Evaluation of model outputs against external data in conjunction with the results from model checking provide information about the structural realism and therefore credibility of the model (Grimm & Berger, 2016).

Model output corroboration

Explanation

Ideally, model outputs or predictions are compared to independent data and patterns that were not used to develop, parameterise, or verify the model. Testing against a dataset of response and predictor variables that are spatially and/or temporally independent from the training dataset minimises the risk of artificially inflating model performance measures (Araújo et al., 2019). Although the corroboration of model outputs against an independent validation dataset is considered the 'gold standard' for showing that a model properly represents the internal organisation of the system, model validation is not always possible because empirical experiments are infeasible or model users are working on rapid-response time-frames, hence, why ecologists often model in the first place (Grimm et al., 2014). Independent predictions might instead be tested on sub-models. Alternatively, patterns in model output that are robust and seem characteristic of the system can be identified and evaluated in consultation with the literature or by experts to judge how accurate the model output is (Grimm et al., 2014).

Preregistration Item

- ☐ State whether you will corroborate the model outputs on external data, and document any independent validation data in step.
- ☐ It is preferable that any independent data used for model evaluation remains unknown to modellers during the process of model building (Dwork et al., 2015), describe the relationship modellers have to model validation data, e.g. will independent datasets be known to any modeller or analyst involved in the model building process?
- ☐ If unable to evaluate the model outputs against independent data, explain why and explain what steps you will take to interrogate the model.

Choose performance metrics and criteria

Explanation

Model performance can be quantified by a range of tests, including measures of agreement between predictions and independent observations, or estimates of accuracy, bias, calibration, discrimination refinement, resolution and skill (Araújo et al., 2019). Note that the performance metrics and criteria in this section are used for evaluating the structured and parameterised models (ideally) on independent holdout data, so this step is additional to any performance criteria used for determining model structure or parameterisation in section X.

Preregistration Item

- ☐ Specify what performance measures you will use to evaluate the model and briefly explain how each test relates to different desired properties of a model's performance.
- ☐ Spatial, temporal and environmental pattern of errors and variance can change the interpretation of model predictions and conservation decisions (Araújo et al., 2019), where relevant and possible, describe how you will characterise and report the spatial, temporal and environmental pattern of errors and variance.
- ☐ If comparing alternative models, specify what measures of model comparison or out-of-sample performance metrics will you use to find support for alternative models or else to optimise predictive ability. State what numerical threshold or qualities you will use for each of these metrics.

Model analysis

Rationale & Explanation

Uncertainty in models arises due to incomplete system understanding (which processes to include, or which interact), from imprecise, finite and sparese data measurements, and from uncertainty in input conditions and scenarios for model simulations or runs (Jakeman et al., 2006). Non-technical uncertainties can also be introduced throughout the modellign process, such as uncertainties arising from issues in problem-framing, indeterminicies, and modeller / client values (Jakeman et al., 2006).
The purpose of model analysis is to prevent blind trust in the model by understanding how model outputs have emerged, and to 'challenge' the model by verifying whether the model is still believable and fit for purpose if one or more parameters are changed

(Grimm et al., 2014).
Model analysis should increase understanding of the model behaviour by identifying which processes and process interactions explain characteristic behaviours of the model system. Model analysis typically consists of sensitivity analyses preceded by uncertainty analyses (Saltelli et al., 2019), and a suite of other simulation or other computational experiments. The aim of such computational experiments is to increase understanding of the model behaviour by identifying which processes and process interactions explain characteristic behaviours of the model system (Grimm et al., 2014). Uncertainty analyses and sensitivity analyses augment one another to draw conclusions about model uncertainty.
Because the results from a full suite of sensitivity analysis and uncertainty analysis can be difficult to interpret due to the number and complexity of causal relations examined (Jakeman et al., 2006), it is useful for the analyst to relate the choice of analysis to the modelling context, purpose and analytical objectives defined in the problem formulation phase, in tandem with any critical uncertainties that have emerged during model development and testing prior to this point.

Uncertainty Analyses

Explanation

Uncertainty can arise from different modelling techniques, response data and predictor variables (Araújo et al., 2019). Uncertainty analyses characterise the uncertainty in model outputs, and identify how uncertainty in model parameters affects uncertainty in model output, but does not identify which model assumptions are driving this behaviour (Grimm et al., 2014; Saltelli et al., 2019). Uncertainty analyses can include propagating known uncertainties through the model, or by investigating the effect of different model scenarios with different parameters and modelling technique combinations (Araújo et al., 2019), for example. It could also include characterising the output distribution, such as through empirical construction using model output data points. It could also include extracting summary statistics like the mean, median and variance from this distribution, and perhaps constructing confidence intervals on the mean (Saltelli et al., 2019).

Preregistration Item

- [ ] Please describe how you will characterise model and data uncertainties, e.g. propagating known uncertainties through the model, investigating the effect of different model scenarios with different parameters and modelling technique combinations (Araújo et al., 2019), or empirically constructing model distributions from model output data points, and extracting summary statistics, including the mean, median, variance, and constructing confidence intervals (Saltelli et al., 2019).

- ☐ Relate your choice of analysis to the context and purposes of the model described in the problem formulation phase. For instance , discrepancies between model output and observed output may be important for forecasting models, where cost, benefit, an risk over a substantial period must be gauged, but much less critical for decision-making or management models where the user may be satisfied with knowing that the predicted ranking order of impacts of alternative scenarios or management options is likely to be correct, with only a rough indication of their sizes" (Jakeman et al., 2006).
- ☐ Briefly describe how you will summarise the results of these in silico experiments with graphical, tabular, or other devices, such as summary statistics.
- ☐ If the chosen modelling approach is able to explicitly articulate uncertainty due to data, measurements or baseline conditions, such as by providing estimates of uncertainty (typically in the form of probabilistic parameter covariance, (Jakeman et al., 2006)), specify which measure of uncertainty you will use.

Sensitivity analyses

Explanation

Sensitivity analysis examines how uncertainty in model outputs can be apportioned to different sources of uncertainty in model input (Saltelli et al., 2019).

Preregistration Item

- ☐ Describe the sensitivity analysis approach you will take: deterministic sensitivity, stochastic sensitivity (variability in the model), or scenario sensitivity (effect of changes based on scenarios).
- ☐ Describe any sensitivity analyses you will conduct by specifying which parameters will be held constant, which will be varied, and the range and intervals of values over which those parameters will be varied.
- ☐ State the primary objective of each sensitivity analysis, for example, to identify which input variables contribute the most to model uncertainty so that these variables can be targeted for further data collection, or alternatively to identify which variables or factors contribute little to overall model outputs, and so can be 'dropped' from future iterations of the model (Saltelli et al., 2019).

Model application or scenario analysis

Preregistration Item

- ☐ Specify any input conditions and relevant parameter values for initial environmental conditions and decision-variables under each scenario specified in section 1.
- ☐ Describe any other relevant technical details of model application, such as methods for how you will implement any simulations or model projections.
- ☐ What raw and transformed model outputs will you extract from the model simulations or projections, and how will you map, plot, or otherwise display and synthesise the results of scenario and model analyses.
- ☐ Explain how you will analyse the outputs to answer your analytical objectives. For instance, describe any trade-off or robustness analyses you will undertake to help evaluate and choose between different alternatives in consultation with experts or decision-makers.

Other simulation experiments / robustness analyses

Preregistration Item

- ☐ Describe any other simulation experiments, robustness analyses or other analyses you will perform on the model, including any metrics and their criteria / thresholds for interpreting the results of the analysis.

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